



Efficacy and safety of cefepime–nacubactam and aztreonam–nacubactam compared with imipenem–cilastatin for complicated urinary tract infection or acute uncomplicated pyelonephritis (Integral-1): a double-blind, randomised phase 3 trial

Satoshi Takahashi, Kazuhiro Tateda, Katsunori Yanagihara, Haihui Huang, Yohei Doi, Mitsuru Yasuda, Kazuaki Matsumoto, Masayo Sumiya, Risako Takaya, Keisuke Suwada, Satoshi Kamiyabu, Yuji Sasagawa, Takeshi Minamida, Seiji Kato, Kenichiro Kondo, Takeshi Naruse, Hiroshige Mikamo

Summary

Background Nacubactam (OP0595) is a newly developed diazabicyclooctane β -lactamase inhibitor used in combination with cefepime or aztreonam. We assessed the efficacy and safety of cefepime–nacubactam and aztreonam–nacubactam versus imipenem–cilastatin in complicated urinary tract infection (cUTI) or acute uncomplicated pyelonephritis.

Methods The Integral-1 global, phase 3, multicentre, randomised, double-blind study recruited adults (aged ≥ 18 years) with cUTI or acute uncomplicated pyelonephritis at 79 sites in Bulgaria, China, Czech Republic, Estonia, Georgia, Japan, Latvia, Lithuania, and Slovakia. Patients were randomly assigned (2:1:1) to receive intravenous cefepime (2 g) plus nacubactam (1 g), aztreonam (2 g) plus nacubactam (1 g), or imipenem (1 g) plus cilastatin (1 g) every 8 h for 5–14 days. Randomisation was stratified by diagnosis and geographical region. The primary endpoint was the proportion of patients achieving composite clinical and microbiological success at test of cure in the microbiological modified intention-to-treat population—all patients who were randomly assigned, received any amount of the study drug, and had a baseline qualifying pathogen that was susceptible to imipenem and meropenem. The prespecified non-inferiority margin was more than 15 percentage points difference; the superiority margin was more than zero percentage points difference, for the lower bound of the two-sided 95% CI for imipenem–cilastatin. Safety was assessed in all patients who received at least one dose of study drug. This study is registered with ClinicalTrials.gov, NCT05887908.

Findings Between May 22, 2023, and Nov 26, 2024, 614 patients were randomly assigned and 431 were included in the primary efficacy analysis (cefepime–nacubactam [$n=214$], aztreonam–nacubactam [$n=112$], or imipenem–cilastatin [$n=105$]); 228 patients (53%) were male and 203 (47%) were female. The primary endpoint was achieved by 176 (82%) of 214, 81 (72%) of 112, and 64 (61%) of 105 patients in the cefepime–nacubactam, aztreonam–nacubactam, and imipenem–cilastatin groups, respectively. The percentage difference in the success rate versus imipenem–cilastatin was 21·3% (95% CI 10·9 to 32·0) for cefepime–nacubactam (non-inferior and superior), and 11·4% (–1·2 to 23·7) for aztreonam–nacubactam (non-inferior). Treatment-emergent adverse events were reported in 100 (33%) of 306, 45 (30%) of 152, and 65 (43%) of 150 patients in the cefepime–nacubactam, aztreonam–nacubactam, and imipenem–cilastatin groups, respectively. No treatment-related deaths occurred.

Interpretation Cefepime–nacubactam and aztreonam–nacubactam are potential treatment options for Gram-negative cUTI and acute uncomplicated pyelonephritis, including infections caused by antimicrobial-resistant strains.

Funding Meiji Seika Pharma and Japan Agency for Medical Research and Development.

Copyright © 2026 Published by Elsevier Ltd. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Introduction

Urinary tract infection (UTI) is a major cause of hospitalisation and is associated with substantial morbidity and mortality and a high economic burden.^{1,2} Complicated UTIs (cUTIs) occur in association with a structural or functional abnormality of the genitourinary tract³ and are associated with a higher risk of treatment

failure, wider spectrum of aetiological bacteria, and a much higher risk of clinical complications compared with uncomplicated UTIs.⁴ UTIs, including cUTIs, are most commonly caused by *Escherichia coli* and *Klebsiella* species, although a wide variety of organisms can cause cUTIs.⁴ Infections with carbapenem-resistant Enterobacterales and third-generation cephalosporin-resistant Enterobacterales

Lancet 2026; 407: 1929–40

See [Comment](#) page 1893

For the plain language summary see Online for appendix 1

Department of Infection Control and Laboratory Medicine, Sapporo Medical University School of Medicine, Sapporo, Japan (S Takahashi MD, M Yasuda MD); Department of Microbiology and Infectious Diseases, Toho University School of Medicine, Tokyo, Japan (K Tateda MD); Department of Laboratory Medicine, Nagasaki University Hospital, Nagasaki, Japan (K Yanagihara MD); Institute of Antibiotics, Huashan Hospital, Fudan University, Shanghai, China (H Huang MD); Departments of Microbiology and Infectious Diseases, Fujita Health University School of Medicine, Toyoake, Japan (Y Doi MD); Division of Infectious Diseases, University of Pittsburgh School of Medicine, Pittsburgh, PA, USA (Y Doi); Division of Pharmacodynamics, Keio University Faculty of Pharmacy, Tokyo, Japan (K Matsumoto PhD); Meiji Seika Pharma, Tokyo, Japan (M Sumiya MSc, R Takaya MPH, K Suwada BSc, S Kamiyabu MSc, Y Sasagawa BSc, T Minamida BSc, S Kato MSc, K Kondo PhD, T Naruse MSc); Department of Clinical Infectious Diseases, Aichi Medical University Hospital, Nagakute, Japan (H Mikamo MD)

Correspondence to:
Dr Hiroshige Mikamo,
Department of Clinical Infectious
Diseases, Aichi Medical
University Hospital, Nagakute,
Aichi 480-1195, Japan
mikamo@aichi-med-u.ac.jp

Research in context

Evidence before this study

To design this study, we searched the Food and Drug Administration, European Medicines Agency (EMA), and Pharmaceuticals and Medical Devices Agency (PMDA) guidelines for information related to conducting clinical trials of complicated urinary tract infections (cUTI). PubMed was also searched for randomised controlled trials intended to support a marketing application for new antibacterial drugs to treat carbapenem-resistant infections that were published between July 10, 2019, and July 17, 2020, using the terms “complicated urinary tract infections”, “phase 2”, and “phase 3”. We also had consultations with the EMA and PMDA regarding the appropriate study population and acceptable non-inferiority margins. To our knowledge, there have been no substantial updates in the evidence regarding trial design guidelines for cUTI or acute uncomplicated pyelonephritis or for similar drug trial designs since 2020.

Nacubactam exhibits potent in-vitro activity against a large group of Enterobacterales when combined with cefepime or aztreonam, including New Delhi metallo- β -lactamase-producing isolates. Human simulated doses of cefepime-nacubactam and aztreonam-nacubactam show potent in-vivo efficacy in mouse thigh and lung infection models. The safety and tolerability of cefepime-nacubactam and aztreonam-nacubactam in healthy adults have been confirmed in phase 1 trials.

Added value of this study

To our knowledge, Integral-1 is the first study to evaluate the efficacy and safety of cefepime-nacubactam and aztreonam-nacubactam in patients with cUTI or acute uncomplicated pyelonephritis. This study showed the efficacy

of cefepime-nacubactam and aztreonam-nacubactam, which was supported by the non-inferiority of cefepime-nacubactam and aztreonam-nacubactam versus imipenem-cilastatin and the superiority of cefepime-nacubactam versus imipenem-cilastatin in patients with cUTI or acute uncomplicated pyelonephritis caused by Gram-negative bacteria except *Acinetobacter* species, including antimicrobial-resistant strains. Cefepime-nacubactam and aztreonam-nacubactam were shown to have acceptable safety profiles. A separate pathogen-directed study in patients with carbapenem-resistant infections will provide additional valuable information about cefepime-nacubactam and aztreonam-nacubactam.

Implications of all the available evidence

Integral-1 showed that cefepime-nacubactam and aztreonam-nacubactam are effective and well tolerated for the treatment of cUTI or acute uncomplicated pyelonephritis and related secondary bacteraemia caused by Gram-negative bacteria except *Acinetobacter* species, including antimicrobial-resistant strains. It was necessary to exclude patients with carbapenem-resistant pathogens, given that imipenem, a carbapenem, was used as the active comparator. Adding the results of this study to those of the existing pharmacokinetic and non-clinical pharmacology studies strengthens the evidence that supports cefepime-nacubactam and aztreonam-nacubactam as important new treatments for antimicrobial-resistant Gram-negative bacterial infections. A clinical trial to assess efficacy of cefepime-nacubactam and aztreonam-nacubactam for infections caused by carbapenem-resistant Enterobacterales (Integral-2) is ongoing.

are difficult to treat because of their resistance to multiple antibacterial agents. Both are listed as critical groups on the WHO Bacterial Priority Pathogens list, representing serious threats to public health.⁵ Infections caused by extended-spectrum β -lactamase (ESBL)-producing bacteria are also difficult to treat because of their resistance to third-generation cephalosporins, often requiring treatment with a carbapenem.⁶ A growing diversity of β -lactamases produced by Gram-negative bacteria poses a challenge to current antibacterial treatment. To help combat such pathogens and prevent the emergence of carbapenem-resistant strains, alternative antibacterials are needed to support carbapenem-sparing treatment strategies.^{3,7-9} The development of new, highly effective β -lactamase inhibitors could contribute to this.

Nacubactam (OP0595) is a recently developed diazabicyclooctane β -lactamase inhibitor with potent activity against carbapenem-resistant Enterobacterales and third-generation cephalosporin-resistant Enterobacterales, including ESBL producers, when combined with cefepime or aztreonam.¹⁰ Nacubactam has a dual mode of action: (1) inhibiting serine

β -lactamases (class A and class C and some in class D), resulting in the protection of the partner β -lactam; and (2) inhibiting penicillin-binding protein 2 of Enterobacterales, resulting in antibacterial activity and enhanced activity of β -lactam agents.¹⁰ Two types of combinations (cefepime-nacubactam and aztreonam-nacubactam) have been developed to adapt to regional differences in the epidemiology of resistant pathogens. Two phase 3 clinical studies have been initiated to further evaluate the clinical use of nacubactam: Integral-1 (NCT05887908), the results of which are reported here, and Integral-2, an ongoing phase 3 study assessing the efficacy and safety of cefepime-nacubactam and aztreonam-nacubactam compared with the best available therapy for adults with various infections, including cUTI and acute uncomplicated pyelonephritis, caused by carbapenem-resistant Enterobacterales (NCT05905055). The Integral-1 study aimed to compare the efficacy and safety of cefepime-nacubactam and aztreonam-nacubactam versus imipenem-cilastatin in patients with cUTI or acute uncomplicated pyelonephritis.

Methods

Study design

Integral-1 is a phase 3, multicentre, randomised, double-blind study. The study was conducted at 79 sites in Bulgaria, China, Czech Republic, Estonia, Georgia, Japan, Latvia, Lithuania, and Slovakia between May 22, 2023, and Nov 26, 2024. A list of principal investigators and study sites is available in appendix 2 (pp 2–5). The protocol was approved by each site's institutional review board or independent ethics committee and is available in appendix 2, along with the statistical analysis plan. The study was conducted in accordance with the Declaration of Helsinki, the International Conference on Harmonisation guidelines for Good Clinical Practice, and applicable laws and regulations. No patients were involved in setting the research questions or the outcome measures, nor were they involved in developing plans for recruitment, design, or implementation of the study. No patients were asked to advise on interpretation or writing up of results.

Patients

Adult patients (aged ≥ 18 years) were eligible to participate in the study if they weighed 140 kg or less, had a diagnosis of cUTI or acute uncomplicated pyelonephritis as defined in appendix 2 (p 6) and an expected intravenous antibacterial treatment duration of at least 5 days for cUTI or acute uncomplicated pyelonephritis, and a known or suspected baseline positive urine culture specimen ($\geq 10^5$ colony-forming units [CFU]/mL of an uncontaminated, Gram-negative uropathogen susceptible to meropenem and imipenem) obtained within 48 h before the first dose of the study drug. Patients with secondary bacteraemia, which was defined as presence of an organism or organisms isolated from a non-contaminated blood culture containing the same organism or organisms isolated from a urine culture, were included in the study, categorised by the primary indication (cUTI or acute uncomplicated pyelonephritis) and recorded as having secondary bacteraemia. The key exclusion criteria were a known imipenem-resistant, or meropenem-resistant, or both, Gram-negative uropathogen ($\geq 10^5$ CFU/mL); detection of only a known Gram-positive uropathogen ($\geq 10^5$ CFU/mL); a contaminated culture (≥ 3 bacterial organisms $\geq 10^5$ CFU/mL isolated from the study-qualifying urine culture); known or suspected single or concurrent infection with *Acinetobacter* species or other organisms not adequately covered by the study drug; a complete obstruction of the urinary tract; receipt of a potentially effective systemic antibacterial therapy within 48 h before the first dose of study drug (with the exception that no more than 15% of patients could have received a single dose of a short-acting antibacterial agent); or an estimated creatinine clearance (CrCl) rate of less than 30 mL/min or more than 240 mL/min. Patients could be enrolled and start intravenous study drug therapy before the study

investigator received the results of the baseline urine culture, in cases where all other entry criteria were fulfilled. If a patient's baseline urine culture was negative after randomisation, the study investigator decided whether the patient should continue to receive the study drug or discontinue the study drug and initiate non-study antibiotics. Sex data were self-reported by the trial participants, with male, female, undifferentiated, and unknown listed as options. All patients provided written informed consent.

See Online for appendix 2

Randomisation and masking

Study investigators enrolled patients at each site. Patients were screened and randomly allocated in a 2:1:1 ratio to receive cefepime–nacubactam, aztreonam–nacubactam, or imipenem–cilastatin using a centralised interactive response technology system. The treatment assignment number and information on assigned treatment group were sent to the unmasked pharmacist, or qualified unmasked designee, who prepared the intravenous study drug at each study site, and a masked randomisation notification was sent to masked study site personnel. To ensure balance among treatment groups, randomisation was stratified by diagnosis (cUTI or acute uncomplicated pyelonephritis) and by geographical region (Japan, China, or other). Enrolment was set so that at least 30% of patients had a diagnosis of acute uncomplicated pyelonephritis and at least 60% had a diagnosis of cUTI. The study investigator, site personnel, sponsor, and sponsor's designees involved in monitoring, data management, or other aspects of the study were unaware of the treatment assignment. The site pharmacist, or qualified designee, who prepared the intravenous study drug was unmasked. The infusion bag containing the study drug was labelled with the patient's identification number but did not identify the study drug; an amber intravenous bag was placed over the infusion bag to maintain masking.

Procedures

Patients were required to be hospitalised at the start of the study and the study drug was administered every 8 h (range 7–9 h) by intravenous infusion over a period of 60 min (range 45–75 min). Patients with a CrCl rate of 30 mL/min or greater and less than 60 mL/min received 1 g cefepime plus 0.5 g nacubactam, 1 g aztreonam plus 0.5 g nacubactam, or 0.5 g imipenem plus 0.5 g cilastatin, every 8 h. Patients with a CrCl rate of 60 mL/min or greater and less than 90 mL/min received 2 g cefepime plus 1 g nacubactam, 2 g aztreonam plus 1 g nacubactam, or 0.75 g imipenem plus 0.75 g cilastatin, every 8 h. Patients with a CrCl rate of 90 mL/min or greater and 240 mL/min or less received 2 g cefepime plus 1 g nacubactam, 2 g aztreonam plus 1 g nacubactam, or 1 g imipenem plus 1 g cilastatin, every 8 h. All patients were treated for a minimum of 5–10 days and up to 14 days if needed. Oral step-down therapy was not permitted.

End of treatment assessments were performed on the day of the last study drug dose or within 24 h following the day of the last study drug dose. Patients attended two follow-up visits, which included a test of cure visit 7 days (range 5–9) after the end of treatment visit and a follow-up visit 14 days (range 12–16) after the end of treatment. At the end of treatment, test of cure, and follow-up visits, patients were assessed for clinical outcome and a urine sample was collected for culture.

Study endpoints

The primary endpoint was the proportion of patients who achieved composite clinical and microbiological success at the test of cure visit in the microbiological modified intention-to-treat (mITT) population. Composite clinical and microbiological success were defined as the composite clinical outcome

of cure and the microbiological outcome of eradication, which were assessed centrally. The clinical outcomes were assessed by the investigators at each study site, and the microbiological outcomes were assessed based on both test results obtained from each study site and those obtained from the central laboratory. Full details of the clinical and microbiological outcome criteria are available in appendix 2 (p 7).

The secondary endpoints included the proportion of patients with composite clinical and microbiological outcome of success, clinical outcome of cure, and microbiological outcome of eradication at the end of treatment and follow-up visits in the microbiological mITT, and at the test of cure visit in patients with secondary bacteraemia at baseline and patients with confirmed ESBL-positive status in the microbiological mITT.

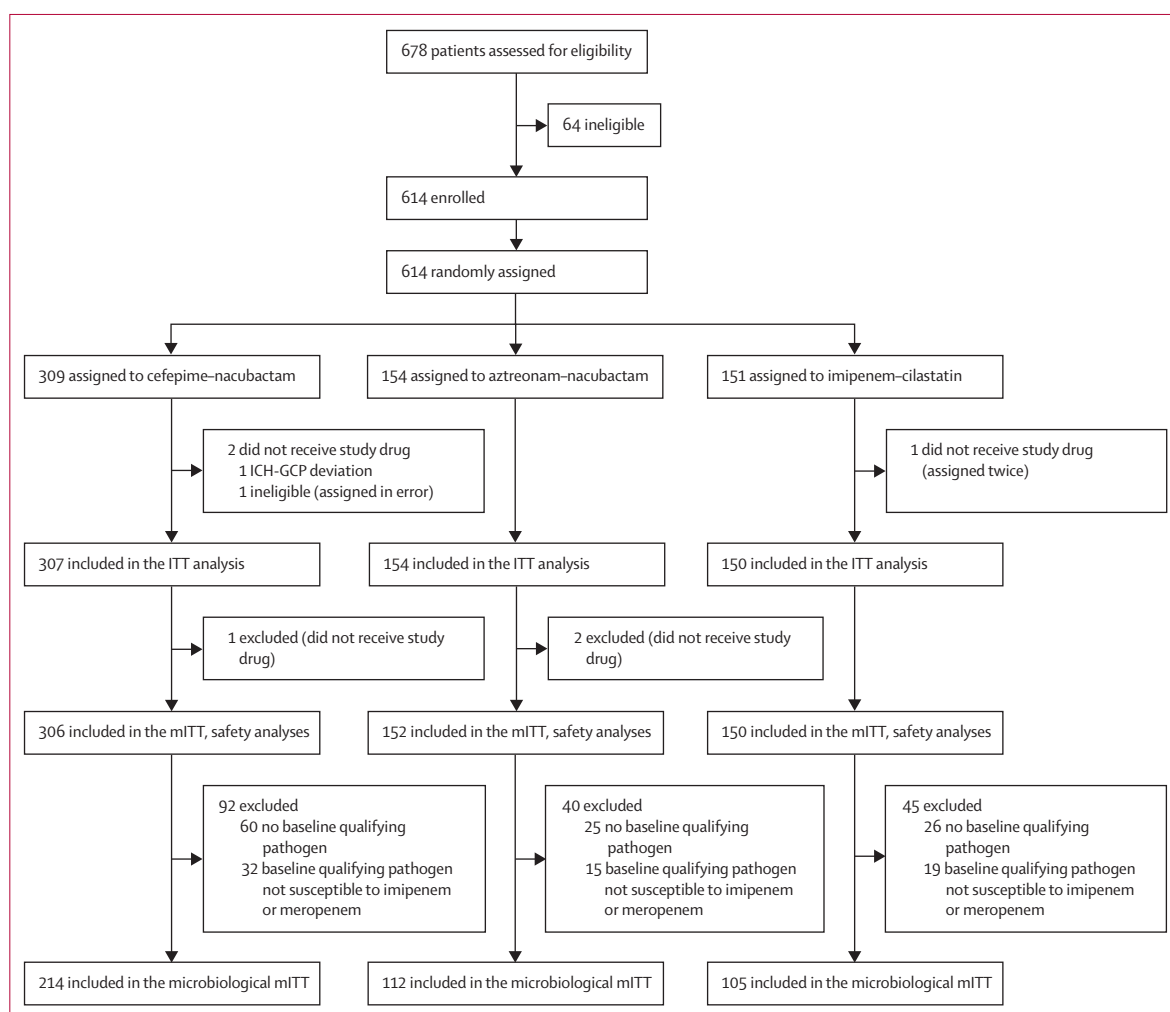


Figure: Trial profile

The ITT population was defined as all patients who were randomly allocated to treatment. The mITT and safety populations were defined as all patients who met the ITT criteria and received any amount of the study drug. The microbiological mITT population was defined as all patients who met the mITT criteria and had a baseline qualifying pathogen that was susceptible to imipenem and meropenem. ICH–GCP=International Council for Harmonisation–Good Clinical Practice. ITT=intention-to-treat. mITT=modified ITT.

Adverse events, treatment-emergent adverse events, and adverse drug reactions were monitored for safety. The study investigator assessed the severity of adverse events and adverse drug reactions and determined the potential relationship of adverse events to the study drug. Adverse events and adverse drug reactions were coded by System Organ Class and Preferred Term according to the Medical Dictionary for Regulatory Activities (MedDRA) version 26.0.

Statistical analysis

Using a 15% non-inferiority margin, a one-sided α of 0.025, 80% power, a composite clinical and microbiological success rate of 70% in each treatment group (cefepime–nacubactam, imipenem–cilastatin) at the test of cure visit, and a 2:1 allocation ratio, a total of 330 patients were required in the microbiological mITT population. Additionally, 110 patients were required in the microbiological mITT population for the aztreonam–nacubactam group in order to evaluate the efficacy and safety of aztreonam–nacubactam with the same precision as the imipenem–cilastatin group. Assuming 75% of patients were evaluable in the microbiological mITT population, it was determined that approximately 600 patients (using a 2:1:1 allocation ratio with 300 patients in the cefepime–nacubactam group and 150 patients each in the aztreonam–nacubactam and imipenem–cilastatin groups) were needed to conduct the study.

The microbiological mITT population included all patients who were randomly assigned, received any amount of the study drug, and had a baseline qualifying pathogen that was susceptible to imipenem and meropenem. The safety population included all patients who received at least one dose of the study drug. Summary statistics were used to describe continuous variables; frequency and percentage were used to describe categorical variables.

For the primary and secondary endpoints, the Miettinen and Nurminen method¹¹ was used to calculate the 95% CIs for the observed difference in the proportion of patients in each group. The prespecified non-inferiority and superiority margins were defined as more than 15 percentage points difference and more than zero percentage points difference, respectively, for the lower bound of the two-sided 95% CI for imipenem–cilastatin. Specifically, if the lower limit of the two-sided 95% CI for the difference between treatment groups at the test of cure visit in the microbiological mITT population was more than 15 percentage points difference for the primary analysis, the null hypothesis was rejected and non-inferiority was concluded based on the primary efficacy endpoint. If non-inferiority was declared for the primary efficacy endpoint, a test for superiority was performed using a hierarchical gatekeeping approach for overall control of the type I error. For this analysis, if the lower bound of the 95% CI for the treatment difference in

treatment success at the test of cure visit in the microbiological mITT population was more than zero, superiority was declared. Furthermore, if non-inferiority of the cefepime–nacubactam group was declared for the primary efficacy endpoint, the non-inferiority hypothesis test for the aztreonam–nacubactam group was performed using a hierarchical gatekeeping approach for overall control of the type I error. For this analysis, and for the difference (aztreonam–nacubactam group minus imipenem–cilastatin group) in treatment success at the test of cure visit, the same approach as in the cefepime–nacubactam group was used.

Patients with missing data or who were lost to follow-up at the test of cure visit were defined as treatment failure

	Cefepime– nacubactam (n=214)	Aztreonam– nacubactam (n=112)	Imipenem– cilastatin (n=105)	Total (n=431)
Age, years	63.62 (14.83)	66.41 (14.04)	65.46 (15.57)	64.79 (14.83)
Sex				
Male	111 (52%)	61 (54%)	56 (53%)	228 (53%)
Female	103 (48%)	51 (46%)	49 (47%)	203 (47%)
Race				
Asian (Japanese)	12 (6%)	6 (5%)	5 (5%)	23 (5%)
Asian (Chinese)	23 (11%)	9 (8%)	7 (7%)	39 (9%)
Asian (other)	0	1 (<1%)	0	1 (<1%)
White	179 (84%)	96 (86%)	93 (89%)	368 (85%)
Geographical region				
Japan	12 (6%)	6 (5%)	5 (5%)	23 (5%)
China	23 (11%)	10 (9%)	7 (7%)	40 (9%)
Other*	179 (84%)	96 (86%)	93 (89%)	368 (85%)
Weight, kg	79.11 (17.75)	81.26 (19.20)	77.47 (17.48)	79.27 (18.08)
Body mass index, kg/m ²	27.52 (5.42)	28.49 (5.37)	26.94 (5.59)	27.62 (5.46)
Primary infection type				
Complicated urinary tract infection	137 (64%)	76 (68%)	73 (70%)	286 (66%)
Acute uncomplicated pyelonephritis	77 (36%)	36 (32%)	32 (30%)	145 (34%)
Creatinine clearance, mL/min				
Total	86.9 (34.9; n=212)	87.8 (40.1)	82.0 (32.4; n=104)	85.9 (35.8; n=428)
<30	1 (<1%)	3 (3%)	0	4 (1%)
30 to <60	46 (21%)	30 (27%)	27 (26%)	103 (24%)
60 to <90	77 (36%)	30 (27%)	44 (42%)	151 (35%)
90 to <240	88 (41%)	49 (44%)	33 (31%)	170 (39%)
Secondary bacteraemia	15 (7%)	9 (8%)	10 (10%)	34 (8%)
Prior short-acting antibacterial therapy	18 (8%)	12 (11%)	12 (11%)	42 (10%)
Resistance type				
ESBL-positive	48 (22%)	34 (30%)	27 (26%)	109 (25%)
<i>Escherichia coli</i>	34 (16%)	25 (22%)	21 (20%)	80 (19%)
<i>Klebsiella pneumoniae</i>	15 (7%)	8 (7%)	6 (6%)	29 (7%)
<i>Proteus mirabilis</i>	0	1 (<1%)	0	1 (<1%)

Data are n (%) or mean (SD). mITT=modified intention-to-treat. ESBL=extended-spectrum β -lactamases. *Includes Bulgaria, Czech Republic, Estonia, Georgia, Latvia, Lithuania, and Slovakia. ESBL defined as testing with a ceftazidime broth microdilution minimum inhibitory concentration of 2 μ g/mL or greater, and a 5 mm or greater increase in zone size when tested with ceftazidime-clavulanate or cefotaxime-clavulanate disks, compared with the zone size for discs containing the agent alone.

Table 1: Patient demographics and background characteristics (microbiological mITT population)

(clinical outcome of failure for the last visit before the test of cure visit) or treatment indeterminate (any clinical outcome except failure for the last visit before the test of cure visit) for the primary endpoint; patients defined as failure or indeterminate were included only in the denominator of the success rate calculation. SAS version 9.4 was used to perform the statistical analysis.

An independent Data Safety Monitoring Board reviewed accumulated safety data and serious adverse events on an ongoing basis and made recommendations based on the safety data. The study was registered with ClinicalTrials.gov (<https://clinicaltrials.gov/study/NCT05887908>).

Role of the funding source

Meiji Seika Pharma had a role in the study design, data collection, data analysis, data interpretation, writing of the report, and decision to submit the paper for publication. The Japan Agency for Medical Research and Development had no role in the study design, data collection, data analysis, data interpretation, writing of the report, and decision to submit the paper for publication.

Results

Between May 22, 2023, and Nov 26, 2024, 614 patients were randomly assigned to receive cefepime–nacubactam (n=309), aztreonam–nacubactam (n=154), or imipenem–cilastatin (n=151; figure). Of the 309 patients assigned to the cefepime–nacubactam group, 306 received the study treatment and were included in the safety population (three patients were excluded: deviation from International Council for Harmonisation—Good Clinical Practice, n=1; ineligible patient randomised in error, n=1; and did not receive study drug, n=1), and 214 were included in the microbiological mITT (92 patients were excluded: no baseline qualifying pathogen, n=60; baseline qualifying pathogen not susceptible to imipenem or meropenem, n=32). Of the 154 patients assigned to the aztreonam–nacubactam group, 152 received the study treatment and were included in the safety population (two patients were excluded: neither received the study drug) and 112 were included in the microbiological mITT population (40 patients were excluded: no baseline qualifying pathogen, n=25; baseline qualifying pathogen not susceptible to imipenem or meropenem, n=15). Of the 151 patients assigned to the imipenem–cilastatin group, 150 received the study treatment and were included in the safety analysis (one patient did not receive the study drug) and 105 were included in the microbiological mITT population (45 patients were excluded: no baseline qualifying pathogen, n=26; baseline qualifying pathogen not susceptible to imipenem or meropenem, n=19).

For the overall microbiological mITT population (n=431), the age, BMI, and CrCl mean rates at baseline were 64.79 years (SD 14.83), 27.62 kg/m² (5.46), and 85.9 mL/min (35.8; n=428); 228 (53%) patients were male and 203 (47%) were female, 368 (85%) were White, 39 (9%) were Chinese, and 23 (5%) were Japanese. Overall of 431 patients, 286 (66%) were diagnosed with cUTI and 145 (34%) were diagnosed with acute uncomplicated pyelonephritis; 34 (8%) had secondary bacteraemia. ESBL-positive pathogens were identified in 109 (25%) of 431 patients. The baseline characteristics and key microbiological features were well balanced among the treatment groups (table 1).

Baseline pathogens were similarly distributed among the treatment groups (table 2). In the overall microbiological mITT population, the most common pathogen was *E coli* (327 [76%] of 431), followed by *K pneumoniae* (60 [14%] of 431). All other pathogens occurred at a frequency of less than 5%. Polymicrobial infection was present in 13 (3%) of 431 patients.

In total, 176 (82%) of 214, 81 (72%) of 112, and 64 (61%) of 105 patients in the cefepime–nacubactam, aztreonam–nacubactam, and imipenem–cilastatin groups, respectively, achieved the primary endpoint of composite clinical outcome and microbiological success at the test of cure (table 3). The percentage difference in the success rate versus the imipenem–cilastatin group

	Cefepime– nacubactam (n=214)	Aztreonam– nacubactam (n=112)	Imipenem– cilastatin (n=105)	Total (n=431)
Baseline pathogen				
<i>Escherichia coli</i>	162 (76%)	89 (80%)	76 (72%)	327 (76%)
<i>Klebsiella pneumoniae</i>	32 (15%)	12 (11%)	16 (15%)	60 (14%)
<i>Proteus mirabilis</i>	2 (1%)	7 (6%)	2 (2%)	11 (3%)
<i>Klebsiella oxytoca</i>	5 (2%)	2 (2%)	1 (1%)	8 (2%)
<i>Pseudomonas aeruginosa</i>	5 (2%)	1 (1%)	2 (2%)	8 (2%)
<i>Enterobacter hormaechei</i>	4 (2%)	1 (1%)	2 (2%)	7 (2%)
<i>Citrobacter freundii</i>	2 (1%)	1 (1%)	1 (1%)	4 (1%)
<i>Klebsiella variicola</i>	2 (1%)	1 (1%)	1 (1%)	4 (1%)
<i>Providencia rettgeri</i>	1 (<1%)	0	2 (2%)	3 (1%)
<i>Citrobacter koseri</i>	1 (<1%)	0	1 (1%)	2 (<1%)
<i>Klebsiella aerogenes</i>	1 (<1%)	1 (1%)	0	2 (<1%)
<i>Proteus hauseri</i>	2 (1%)	0	0	2 (<1%)
<i>Acinetobacter junii</i>	0	1 (1%)	0	1 (<1%)
<i>Citrobacter spp</i>	0	0	1 (1%)	1 (<1%)
<i>Enterobacter spp</i>	0	0	1 (1%)	1 (<1%)
<i>Enterobacter bugandensis</i>	1 (<1%)	0	0	1 (<1%)
<i>Raoultella ornithinolytica</i>	1 (<1%)	0	0	1 (<1%)
<i>Serratia marcescens</i>	1 (<1%)	0	0	1 (<1%)
Polymicrobial infection	8 (4%)	4 (4%)	1 (1%)	13 (3%)
<i>Escherichia coli</i> and <i>Klebsiella pneumoniae</i>	4 (2%)	0	0	4 (1%)
<i>Escherichia coli</i> and <i>Pseudomonas aeruginosa</i>	2 (1%)	1 (1%)	0	3 (1%)
<i>Escherichia coli</i> and <i>Proteus mirabilis</i>	1 (<1%)	1 (1%)	0	2 (<1%)
<i>Klebsiella oxytoca</i> and <i>Klebsiella pneumoniae</i>	0	2 (2%)	0	2 (<1%)
<i>Klebsiella pneumoniae</i> and <i>Providencia rettgeri</i>	1 (<1%)	0	1 (1%)	2 (<1%)

mITT=modified intention-to-treat.

Table 2: Summary of baseline pathogens (microbiological mITT population)

was 21.3% (95% CI 10.9 to 32.0) for the cefepime–nacubactam group, meeting both the superiority and non-inferiority margins, and 11.4% (–1.2 to 23.7) for the aztreonam–nacubactam group, meeting the non-inferiority margin. The clinical outcome of cure was achieved for 195 (91%) of 214, 103 (92%) of 112, and 92 (88%) of 105 patients in the cefepime–nacubactam, aztreonam–nacubactam, and imipenem–cilastatin groups, respectively (table 3). The percentage differences in the cure rate for the cefepime–nacubactam and aztreonam–nacubactam groups versus the imipenem–cilastatin group were 3.5% (95% CI –3.3 to 11.8) and 4.3% (–3.9 to 13.0), respectively. The microbiological outcome of eradication was achieved for 184 (86%) of 214, 83 (74%) of 112, and 67 (64%) of 105 patients in the cefepime–nacubactam, aztreonam–nacubactam, and imipenem–cilastatin groups, respectively (table 3). The percentage differences in the eradication rate for the cefepime–nacubactam and aztreonam–nacubactam groups versus the imipenem–cilastatin group were 22.2% (12.2 to 32.7) and 10.3% (–2.0 to 22.5), respectively. Risk differences according to sex, age, and diagnosis at baseline entry are shown in appendix 2 (p 8).

At the end of treatment, the success rates were 93% (199 of 214), 91% (102 of 112), and 90% (94 of 105) for the cefepime–nacubactam, aztreonam–nacubactam, and imipenem–cilastatin groups, respectively (difference: cefepime–nacubactam vs imipenem–cilastatin, 3.5% [95% CI –2.7 to 11.3]; aztreonam–nacubactam vs imipenem–cilastatin group, 1.5% [95% CI –6.6 to 10.0]; table 4). The respective cure rates were 95% (204 of 214), 95% (106 of 112), and 95% (100 of 105), and eradication rates were 95% (203 of 214), 94% (105 of 112), and 90%

(95 of 105). At the follow-up visit, the success rates were 69% (147 of 214), 65% (73 of 112), and 62% (65 of 105) for the cefepime–nacubactam, aztreonam–nacubactam, and imipenem–cilastatin groups, respectively (difference: cefepime–nacubactam vs imipenem–cilastatin, 6.8% [95% CI –4.1 to 18.1]; aztreonam–nacubactam vs imipenem–cilastatin group, 3.3% [95% CI –9.5 to 16.0]; table 4). The respective cure rates were 88% (188 of 214), 88% (98 of 112), and 85% (89 of 105), and eradication rates were 73% (157 of 214), 67% (75 of 112), and 65% (68 of 105). The sensitivity analysis (adjusted for the randomisation stratification factors of baseline entry diagnosis [cUTI or acute uncomplicated pyelonephritis] and geographical region [Japan, China, or other]) indicated consistent results with the primary analysis (table 3) and confirmed the overall superiority of cefepime–nacubactam (176 [82%] of 214 patients) and non-inferiority of aztreonam–nacubactam (81 [72%] of 112 patients) to imipenem–cilastatin (64 [61%] of 105 patients) for the composite clinical and microbiological success at the test of cure visit with treatment differences of 21.3% (95% CI 12.1 to 32.0) and 11.4% (95% CI –0.1 to 23.5), respectively (appendix 2 pp 9–10).

Among patients with secondary bacteraemia, success rates for clinical cure and microbiological eradication for secondary bacteraemia at the test of cure were 80% (12 of 15), 89% (eight of nine), and 60% (six of ten) for the cefepime–nacubactam, aztreonam–nacubactam, and imipenem–cilastatin groups, respectively (treatment difference: cefepime–nacubactam vs imipenem–cilastatin, 20.0% [95% CI –16.0 to 54.3]; aztreonam–nacubactam vs imipenem–cilastatin group, 28.9% [–13.0 to 62.1]; appendix 2 pp 11–12). Also among patients with secondary bacteraemia, 11 (73%) of 15,

	Cefepime–nacubactam (n=214)	Aztreonam–nacubactam (n=112)	Imipenem–cilastatin (n=105)	Treatment difference (95% CI)	
				Cefepime–nacubactam vs imipenem–cilastatin	Aztreonam– nacubactam vs imipenem–cilastatin
Composite clinical and microbiological success					
Success	176 (82.2% [76.5 to 87.1])	81 (72.3% [63.1 to 80.4])	64 (61.0% [50.9 to 70.3])	21.3% (10.9 to 32.0)	11.4% (–1.2 to 23.7)
Failure	26 (12%)	27 (24%)	34 (32%)	..	..
Indeterminate	12 (6%)	4 (4%)	7 (7%)	..	..
Clinical outcome					
Cure	195 (91.1% [86.5 to 94.6])	103 (92.0% [85.3 to 96.3])	92 (87.6% [79.8 to 93.2])	3.5% (–3.3 to 11.8)	4.3% (–3.9 to 13.0)
Failure	7 (3%)	3 (3%)	5 (5%)	..	..
Indeterminate	12 (6%)	6 (5%)	8 (8%)	..	..
Microbiological outcome					
Eradication	184 (86.0% [80.6 to 90.3])	83 (74.1% [65.0 to 81.9])	67 (63.8% [53.9 to 73.0])	22.2% (12.2 to 32.7)	10.3% (–2.0 to 22.5)
Persistence	0	0	0	..	..
Recurrence	14 (7%)	22 (20%)	28 (27%)	..	..
Indeterminate	16 (7%)	7 (6%)	10 (10%)	..	..

Data are n (% [95% CI]). mITT=modified intention-to-treat.

Table 3: Clinical and microbiological success rates and outcomes at test of cure visit and between-group comparisons (mITT)

	Cefepime– nacubactam (n=214)	Aztreonam– nacubactam (n=112)	Imipenem– cilastatin (n=105)	Treatment difference (95% CI)	
				Cefepime– nacubactam vs imipenem– cilastatin	Aztreonam– nacubactam vs imipenem– cilastatin
End of treatment					
Composite clinical and microbiological success					
Success	199 (93.0% [88.7 to 96.0])	102 (91.1% [84.2 to 95.6])	94 (89.5% [82.0 to 94.7])	3.5% (–2.7 to 11.3)	1.5% (–6.6 to 10.0)
Failure	10 (5%)	5 (4%)	7 (7%)	..	..
Indeterminate	5 (2%)	5 (4%)	4 (4%)	..	..
Clinical outcome					
Cure	204 (95.3% [91.6 to 97.7])	106 (94.6% [88.7 to 98.0])	100 (95.2% [89.2 to 98.4])	0.1% (–4.6 to 6.4)	–0.6% (–7.1 to 6.0)
Failure	5 (2%)	1 (1%)	1 (1%)	..	..
Indeterminate	5 (2%)	5 (4%)	4 (4%)	..	..
Microbiological outcome					
Eradication	203 (94.9% [91.0 to 97.4])	105 (93.8% [87.5 to 97.5])	95 (90.5% [83.2 to 95.3])	4.4% (–1.3 to 11.9)	3.3% (–4.2 to 11.2)
Persistence	0	0	0	..	..
Recurrence	0	3 (3%)	1 (1%)	..	..
Indeterminate	11 (5%)	4 (4%)	9 (9%)	..	..
Follow-up					
Composite clinical and microbiological success					
Success	147 (68.7% [62.0 to 74.8])	73 (65.2% [55.6 to 73.9])	65 (61.9% [51.9 to 71.2])	6.8% (–4.1 to 18.1)	3.3% (–9.5 to 16.0)
Failure	54 (25%)	34 (30%)	34 (32%)	..	..
Indeterminate	13 (6%)	5 (4%)	6 (6%)	..	..
Clinical outcome					
Cure	188 (87.9% [82.7 to 91.9])	98 (87.5% [79.9 to 93.0])	89 (84.8% [76.4 to 91.0])	3.1% (–4.5 to 12.1)	2.7% (–6.6 to 12.3)
Failure	7 (3%)	4 (4%)	2 (2%)	..	..
Recurrence	6 (3%)	5 (4%)	7 (7%)	..	..
Indeterminate	13 (6%)	5 (4%)	7 (7%)	..	..
Microbiological outcome					
Eradication*	157 (73.4% [66.9 to 79.2])	75 (67.0% [57.4 to 75.6])	68 (64.8% [54.8 to 73.8])	8.6% (–2.0 to 19.7)	2.2% (–10.4 to 14.8)
Presumed eradication†	56 (26%)	30 (27%)	18 (17%)	..	..
Persistence	0	0	1 (1%)	..	..
Recurrence	40 (19%)	29 (26%)	25 (24%)	..	..
Indeterminate	17 (8%)	8 (7%)	11 (10%)	..	..

Data are n (%) [95% CI]. CFU= colony-forming units. mITT=modified intention-to-treat. *Includes presumed eradication. †Presumed eradication was assessed at follow-up and categorised as eradication only if the baseline qualifying Gram-negative pathogen was eradicated (reduced to <10³ CFU/mL) at the test of cure visit, and a urine sample could not be collected but there was a clinical outcome of cure at follow-up; therefore, a urine sample should also have been collected at follow-up, unless the baseline qualifying Gram-negative pathogen was eradicated (reduced to <10³ CFU/mL) at the treatment of cure visit.

Table 4: Clinical and microbiological success rates and outcomes at end of treatment visit and follow-up visit and between-group comparisons (microbiological mITT population)

four (44%) of nine, and two (20%) of ten patients in the cefepime–nacubactam, aztreonam–nacubactam, and imipenem–cilastatin groups, respectively, achieved composite clinical outcome and microbiological success from cUTI or acute uncomplicated pyelonephritis at the

test of cure (treatment difference: cefepime–nacubactam vs imipenem–cilastatin 53.3% [95% CI 12.8 to 78.1]; aztreonam–nacubactam vs imipenem–cilastatin group, 24.4% [–18.4 to 60.5]). The respective cure rates were 80% (12 of 15), 89% (eight of nine), and 50% (five of ten), and eradication rates were 73% (11 of 15), 44% (four of nine), and 30% (three of ten) in the cefepime–nacubactam, aztreonam–nacubactam, and imipenem–cilastatin groups, respectively.

Among patients with ESBL-producing bacterial infections, success rates were 75% (36 of 48), 68% (23 of 34), and 56% (15 of 27) at the test of cure for the cefepime–nacubactam, aztreonam–nacubactam, and imipenem–cilastatin groups, respectively (treatment difference: cefepime–nacubactam vs imipenem–cilastatin, 19.4% [95% CI –2.6 to 41.1]; aztreonam–nacubactam vs imipenem–cilastatin group, 12.1% [–12.3 to 35.7]; appendix 2 pp 13–14). The respective cure rates were 85% (41 of 48), 91% (31 of 34), and 85% (23 of 27), and eradication rates were 81% (39 of 48), 68% (23 of 34), and 56% (15 of 27; appendix 2 pp 13–14).

In the cefepime–nacubactam, aztreonam–nacubactam, and imipenem–cilastatin groups 102 (33%) of 306, 47 (31%) of 152, and 66 (44%) of 150 patients experienced any adverse event, respectively; six (2%) of 306, four (3%) of 152, and five (3%) of 150 patients experienced a serious adverse event; and one adverse event leading to death (lung cancer progression, not related to the study drug or intravenous procedure) was reported in the cefepime–nacubactam group (table 5). In all three groups, most treatment-emergent adverse events and adverse drug reactions were mild or moderate in severity. Treatment-emergent adverse events leading to the discontinuation of the study drug occurred in seven (2%) of 306, two (1%) of 152, and two (1%) of 150 patients in the cefepime–nacubactam, aztreonam–nacubactam, and imipenem–cilastatin groups, respectively; and treatment-emergent adverse events leading to study discontinuation occurred in seven (2%) of 306, two (1%) of 152, and one (1%) of 150 patients (table 5). The most common treatment-emergent adverse events by MedDRA preferred term were headache, which occurred most frequently in the aztreonam–nacubactam group (5% [eight of 152]), followed by the imipenem–cilastatin group (4% [six of 150]) and the cefepime–nacubactam group (3% [ten of 306]); diarrhoea, which occurred most frequently in the cefepime–nacubactam group (5% [14 of 306]), followed by the imipenem–cilastatin group (4% [six of 150]) and the aztreonam–nacubactam group (2% [three of 152]); and nausea, which occurred most frequently in the imipenem–cilastatin group (7% [ten of 150]), followed by the cefepime–nacubactam group (2% [five of 306]) and the aztreonam–nacubactam group (1% [one of 152]; table 5).

In the cefepime–nacubactam, aztreonam–nacubactam, and imipenem–cilastatin groups, 44 (14%) of 306,

	Cefepime–nacubactam (n=306)	Aztreonam– nacubactam (n=152)	Imipenem–cilastatin (n=150)	Total (n=608)
Any adverse event	102 (33% [208])	47 (31% [105])	66 (44% [135])	215 (35% [448])
Any serious adverse event	6 (2% [7])	4 (3% [4])	5 (3% [7])	15 (2% [18])
Any adverse event leading to death	1 (<1% [1])	0	0	1 (<1% [1])
Any treatment-emergent adverse event	100 (33% [203])	45 (30% [99])	65 (43% [132])	210 (35% [434])
Mild	67 (22%)	35 (23%)	42 (28%)	144 (24%)
Moderate	28 (9%)	9 (6%)	20 (13%)	57 (9%)
Severe	5 (2%)	1 (1%)	3 (2%)	9 (1%)
Any adverse drug reaction	44 (14% [58])	15 (10% [29])	27 (18% [40])	86 (14% [127])
Mild	35 (11%)	14 (9%)	19 (13%)	68 (11%)
Moderate	8 (3%)	0	7 (5%)	15 (2%)
Severe	1 (<1%)	1 (1%)	1 (1%)	3 (<1%)
Any intravenous procedure-related treatment-emergent adverse event	14 (5% [15])	5 (3% [6])	8 (5% [12])	27 (4% [33])
Any serious treatment-emergent adverse event	6 (2% [7])	4 (3% [4])	5 (3% [7])	15 (2% [18])
Any treatment-emergent adverse event leading to discontinuation of study drug	7 (2% [10])	2 (1% [2])	2 (1% [3])	11 (2% [15])
Any treatment-emergent adverse event leading to discontinuation of study	7 (2% [7])	2 (1% [2])	1 (1% [2])	10 (2% [11])
Treatment-emergent adverse events by System Organ Class and MedDRA preferred term (experienced by ≥2% of patients in any treatment group)				
Gastrointestinal disorders	32 (10% [37])	13 (9% [18])	28 (19% [36])	73 (12% [91])
Diarrhoea	14 (5% [14])	3 (2% [3])	6 (4% [6])	23 (4% [23])
Nausea	5 (2% [5])	1 (1% [1])	10 (7% [10])	16 (3% [16])
Constipation	6 (2% [6])	3 (2% [3])	4 (3% [6])	13 (2% [15])
Vomiting	2 (<1% [2])	1 (1% [1])	5 (3% [5])	8 (1% [8])
Abdominal pain upper	1 (<1% [1])	1 (1% [1])	3 (2% [3])	5 (1% [5])
Infections and infestations	18 (6% [18])	14 (9% [14])	16 (11% [20])	48 (8% [52])
Asymptomatic bacteriuria	3 (1% [3])	7 (5% [7])	5 (3% [5])	15 (2% [15])
Nervous system disorders*	16 (5% [18])	9 (6% [9])	9 (6% [10])	34 (6% [37])
Headache	10 (3% [11])	8 (5% [8])	6 (4% [6])	24 (4% [25])
General disorder and administration site conditions	8 (3% [8])	5 (3% [5])	12 (8% [17])	25 (4% [30])
Pyrexia	4 (1% [4])	2 (1% [2])	3 (2% [3])	9 (1% [9])
Metabolism and nutrition disorders	11 (4% [17])	4 (3% [4])	4 (3% [8])	19 (3% [31])
Hypokalaemia	6 (2% [6])	1 (1% [1])	2 (1% [2])	9 (1% [9])
Skin and subcutaneous tissue disorders	8 (3% [8])	4 (3% [4])	4 (3% [4])	16 (3% [16])
Rash	2 (<1% [2])	2 (1% [2])	4 (3% [4])	8 (1% [8])
Vascular disorders	11 (4% [11])	5 (3% [5])	0	16 (3% [16])
Hypertension	5 (2% [5])	3 (2% [3])	0	8 (1% [8])

Data are n, participants (% [n, events]). Events are coded by System Organ Class and preferred term according to MedDRA version 26.0. MedDRA=Medical Dictionary for Regulatory Activities. *All nervous system disorders other than headache occurred at a frequency of less than 2%: cefepime–nacubactam, dizziness (three treatment-emergent adverse events in three patients [1%]), and cervical radiculopathy, dizziness postural, encephalopathy, and radiculopathy (one treatment-emergent adverse event in one patient [<1%] each); aztreonam–nacubactam, cerebral infarction (one treatment-emergent adverse event in one patient [<1%]); imipenem–cilastatin, neuralgia, paraesthesia, presyncope, and tremor (one treatment-emergent adverse event in one patient [<1%] each).

Table 5: Overall summary of adverse events, treatment-emergent adverse events, adverse drug reactions, and breakdown of treatment-emergent adverse events occurring in minimum 2% of patients in any treatment group by System Organ Class and MedDRA preferred term (safety population)

15 (10%) of 152, and 27 (18%) of 150 patients experienced any study drug-related treatment-emergent adverse event, respectively (appendix 2 pp 15–16).

Discussion

Our results show the non-inferiority of cefepime–nacubactam and aztreonam–nacubactam versus imipenem–cilastatin and the superiority of cefepime–nacubactam versus imipenem–cilastatin for the

treatment of patients hospitalised with cUTI or acute uncomplicated pyelonephritis. The β -lactam plus nacubactam combinations were effective in patients with secondary bacteraemia, and efficacy in patients with ESBL-producing bacterial infections was demonstrated in both nacubactam groups. Cefepime–nacubactam and aztreonam–nacubactam were well tolerated and no new safety concerns were identified. The choice of imipenem–cilastatin as the comparator used in this study was based

on its demonstrated efficacy against Gram-negative pathogens isolated from patients with cUTI or acute uncomplicated pyelonephritis. The drug combination is widely used to treat cUTIs, and carbapenems are the drug of choice against ESBL-producing Gram-negative pathogens, in particular for serious infections. Additionally, imipenem–cilastatin is available for the treatment of cUTI and acute uncomplicated pyelonephritis in both Europe and the Asia–Pacific region.^{12,13} Because baseline characteristics and microbiological features were well balanced between treatment groups overall and in the primary analysis population, the superiority of cefepime–nacubactam was not explained by differences in these factors across treatment groups.

Several antibacterials have been evaluated in clinical studies for treating cUTI and acute uncomplicated pyelonephritis. These include cefepime–taniborbactam (a β -lactam and β -lactamase inhibitor combination);¹⁴ cefiderocol (a siderophore cephalosporin antibacterial);¹⁵ a combination of relebactam (a β -lactamase inhibitor that can restore imipenem activity against imipenem-non-susceptible pathogens) with imipenem–cilastatin;¹⁶ and ceftazidime–avibactam.¹⁷ Avibactam is a β -lactamase inhibitor that restores the in-vitro activity of ceftazidime against several β -lactamase-producing bacteria.¹⁸ Carbapenems have had inferior outcomes compared with newer drugs in clinical trials of cUTI.^{14,19}

Strengths of our study include its double-blind, comparative phase 3 design, which provides a high level of evidence; the use of a composite primary outcome of clinical cure and microbiological eradication plus a more stringent definition of microbiologic eradication ($<10^3$ CFU/mL on urine culture), which is stricter than the primary outcomes used in registrational trials of recently approved treatments for cUTI (eg, cefiderocol,¹⁵ relebactam,¹⁶ and ceftazidime–avibactam¹⁷); and the ability of the study to show the superiority of cefepime–nacubactam versus imipenem–cilastatin for the treatment of patients hospitalised with cUTI or acute uncomplicated pyelonephritis.

We suggest that treatment differences observed in the results of the primary composite efficacy endpoint in the microbiological mITT group are similar to differences in efficacy in cases of ESBL-producing bacteria and secondary bacteraemia. Future studies include efficacy analyses in patients with baseline qualifying pathogens that were not susceptible to imipenem or meropenem in the present study and the ongoing phase 3 Integral-2 study (NCT05905055), which will assess the efficacy and safety of cefepime–nacubactam and aztreonam–nacubactam compared with the best available therapy for adults with infections caused by carbapenem-resistant Enterobacterales.

Regarding safety, the incidences of treatment-emergent adverse events and study drug-related treatment-emergent adverse events were lower in the

cefepime–nacubactam group (33% and 14%, respectively) and aztreonam–nacubactam group (30% and 10%, respectively) compared with the imipenem–cilastatin group (43% and 18%, respectively). The incidences of study drug-related treatment-emergent adverse events observed for cefepime–nacubactam and aztreonam–nacubactam were similar to those reported for cefepime alone or aztreonam alone, based on their respective package inserts;^{20,21} therefore, our results suggest that nacubactam added to either drug does not alter the safety profile.

This study has some limitations to consider. Patients with carbapenem-resistant infections were excluded from this study, as imipenem (a carbapenem class antibiotic) with cilastatin was used as the control treatment. By excluding the oral step-down process in this trial, the effects of the study treatments could be assessed without the potential confounding influence of additional antibacterial therapy. This strategy has also been adopted in previous similar studies.^{14,15} Nevertheless, we note that the absence of a step-down process to oral medication and the need for inpatient participation in this study might not reflect the situation in real-world clinical practice. To investigate efficacy and safety with the same precision as in the imipenem–cilastatin group, the aztreonam–nacubactam group was assigned the same target sample size as the imipenem–cilastatin group. Because the target sample size of the aztreonam–nacubactam group was half that of the cefepime–nacubactam group, the power to detect superiority of aztreonam–nacubactam over imipenem–cilastatin was insufficient, and superiority of aztreonam–nacubactam over imipenem–cilastatin was not included in the hypotheses to be tested in this study. It was therefore not possible to consider the superiority of aztreonam–nacubactam. Most patients in this study (85% of the microbiological mITT population) were located in Eastern Europe. Although it is possible that regional differences in susceptibility patterns might exist, geographical location did not alter either the pathophysiological features of cUTI or acute uncomplicated pyelonephritis or the expected response to antibacterials, as inclusion in the primary analysis population required that all study drugs were active against the baseline pathogens. Moreover, the pharmacokinetic profiles are known to be similar between White and Asian (Japanese or Chinese) individuals. Pharmacokinetic data (area under the concentration–time curve) for White versus Asian populations have been described as follows: 1 g nacubactam, 118 $\mu\text{g}\cdot\text{h/mL}$ ²² versus 129 $\mu\text{g}\cdot\text{h/mL}$;²³ 2 g cefepime, 285 $\mu\text{g}\cdot\text{h/mL}$ ²⁴ versus 289 $\mu\text{g}\cdot\text{h/mL}$;²⁵ and 2 g aztreonam, 294–469 $\mu\text{g}\cdot\text{h/mL}$ ²⁶ versus 389 $\mu\text{g}\cdot\text{h/mL}$.²⁷ Finally, although genomic characterisation of the resistance mechanisms was performed, the results of this analysis are planned to be reported in a future publication.

In conclusion, the composite and microbiological success rate of cefepime–nacubactam at the test of cure visit showed non-inferiority and superiority versus imipenem–cilastatin and that of aztreonam–nacubactam showed non-inferiority versus imipenem–cilastatin. Cefepime–nacubactam and aztreonam–nacubactam were well tolerated and no new safety concerns were identified. Thus, cefepime–nacubactam and aztreonam–nacubactam are potential treatment options for patients with cUTI and acute uncomplicated pyelonephritis caused by Gram-negative bacteria except *Acinetobacter* species, including infections caused by antimicrobial-resistant strains.

Contributors

ST, KT, KY, YD, MY, KM, MS, RT, SeK, KK, and HM participated in the conception and design of the study and contributed to the writing of the protocol. KY and HH participated in the study as investigators. ST, MS, RT, KS, SaK, TM, SeK, KK, and HM contributed to study oversight. YS performed all statistical elements of the study design and analysis of the results. SeK and KK were responsible for project administration. TN contributed to the funding acquisition. All authors contributed to the interpretation of the results. MS led the writing of the initial drafts of the manuscript on which all authors commented, and all authors agreed with the decision to submit the final manuscript for publication. ST and KS directly accessed and verified the underlying data, and were responsible for the decision to submit for publication.

Declaration of interests

ST reports grants from Meiji Seika Pharma and Shimadzu; consulting fees from Meiji Seika Pharma, Hologic Japan, MSD, Nippon Shinyaku, and Fuji Film Toyama Chemical; honoraria from Meiji Seika Pharma, Shionogi, MSD, Sekisui Medical, Pfizer, Roche Diagnostics, Abbott Japan, Denka, Sysmex, Asahi Kasei, Fuji Rebio, and Gilead Sciences; payment for expert testimony from GSK; leadership of fiduciary role for LSI Medience. KT reports grants, consulting fees, and honoraria from Meiji Seika Pharma. KY reports grants from Meiji Seika Pharma, consulting fees and honoraria from Meiji Seika Pharma, Shionogi, Daiichi Sankyo Company, Merck Sharp and Dohme, Becton, Dickinson and Company Japan, BioMérieux Japan, Fujifilm Toyama Chemical, GlaxoSmithKline, Kyorin Pharmaceutical, Pfizer, GramEye, Roche Diagnostics, and Eiken Chemical. HH reports grants from Meiji Seika. YD reports consulting fees from Meiji Seika Pharma, Shionogi, and GSK; honoraria from Shionogi, Gilead, BD, and Chugai. MY reports grants from the Japan Agency for Medical Research and Development; Health Labour Sciences Research, and Grant-in-aid for Scientific Research; consulting fees from Meiji Seika Pharma; honoraria from Kyorin Pharmaceutical, Medical & Biological Laboratories, Shionogi, Kyokuto Pharmaceutical Industrial, Sysmex Corporation, Toyobo, and GSK; payment for expert testimony from GSK. KM reports grants from Meiji Seika Pharma, Sumitomo Pharma, Kyorin Pharmaceutical, Earth Corporation, Kenei Pharmaceutical, and Shionogi; and consulting fees and honoraria from Meiji Seika Pharma. MS, RT, KS, SaK, YS, TM, SeK, KK, and TN report full-time employment and support for attending meetings or travel funded by Meiji Seika Pharma. TN reports board membership at Meiji Seika Pharma. HM reports grants from Meiji Seika Pharma, Miyarisan Pharmaceutical, and Kyorin Pharmaceutical; consulting fees from Meiji Seika Pharma; and honoraria from Meiji Seika Pharma, Miyarisan Pharmaceutical, Shionogi, Pfizer Japan, GlaxoSmithKline, Daiichi Sankyo, MSD, Kyorin Pharmaceutical, Gilead Sciences, Sumitomo Pharma, Asahi Kasei Pharma Corporation, Fukoku, Tsumura, Nipro Corporation, and Roche Diagnostics.

Data sharing

The data generated in this study will be provided upon reasonable request to the authors at clinical-trials@meiji.com after publication of this paper. Data sharing, including the scope of sharing, will be conducted with researchers approved for data use through internal

evaluation and for the purpose of achieving the objectives of approved proposals. A data use agreement must be concluded.

Acknowledgments

The authors thank all investigators and patients involved in this clinical trial programme; Sarah Bubeck, of Edanz (www.edanz.com) for providing medical writing support, which was funded by Meiji Seika Pharma, in accordance with Good Publication Practice guidelines (<https://www.ismpp.org/gpp-2022>); and Marito Araki, of Meiji Seika Pharma, and Hayato Okade, of Meiji Seika Pharma, for providing additional writing support. This research was funded by Meiji Seika Pharma, and the Japan Agency for Medical Research and Development.

References

- Zeng Z, Zhan J, Zhang K, Chen H, Cheng S. Global, regional, and national burden of urinary tract infections from 1990 to 2019: an analysis of the global burden of disease study 2019. *World J Urol* 2022; **40**: 755–63.
- Foxman B. Epidemiology of urinary tract infections: incidence, morbidity, and economic costs. *Dis Mon* 2003; **49**: 53–70.
- Tamma PD, Heil EL, Justo JA, Mathers AJ, Satlin MJ, Bonomo RA. Infectious Diseases Society of America 2024 guidance on the treatment of antimicrobial-resistant Gram-negative infections. *Clin Infect Dis* 2024; published online Aug 7. <http://doi.org/10.1093/cid/ciae403>.
- Sabih A, Leslie SW. Complicated urinary tract infections. 2024. StatPearls Publishing, 2025.
- WHO. WHO bacterial priority pathogens list, 2024. 2024. <https://iris.who.int/bitstream/handle/10665/376776/9789240093461-eng.pdf?sequence=1> (accessed Sept 29, 2025).
- Yoon YK, Kim JH, Sohn JW, Yang KS, Kim MJ. Role of piperacillin/tazobactam as a carbapenem-sparing antibiotic for treatment of acute pyelonephritis due to extended-spectrum β -lactamase-producing *Escherichia coli*. *Int J Antimicrob Agents* 2017; **49**: 410–15.
- Wilson APR. Sparing carbapenem usage. *J Antimicrob Chemother* 2017; **72**: 2410–17.
- Karakos I, Giamarellou H. Carbapenem-sparing strategies for ESBL producers: when and how. *Antibiotics (Basel)* 2020; **9**: 61.
- Paul M, Carrara E, Retamar P, et al. European Society of Clinical Microbiology and Infectious Diseases (ESCMID) guidelines for the treatment of infections caused by multidrug-resistant Gram-negative bacilli (endorsed by European society of intensive care medicine). *Clin Microbiol Infect* 2022; **28**: 521–47.
- Morinaka A, Tsutsumi Y, Yamada M, et al. OP0595, a new diazabicyclooctane: mode of action as a serine β -lactamase inhibitor, antibiotic and β -lactam ‘enhancer’. *J Antimicrob Chemother* 2015; **70**: 2779–86.
- Miettinen O, Nurminen M. Comparative analysis of two rates. *Stat Med* 1985; **4**: 213–26.
- Bonkat G, Bartoletti R, Bruyère F, et al. EAU guidelines on urological infections. 2020. <https://d56bochluxqnz.cloudfront.net/documents/EAU-Guidelines-on-Urological-infections-2020.pdf> (accessed Sept 29, 2025).
- Japanese Association for Infectious Disease/Japanese Society of Chemotherapy, JAID/JSC Guide/Guidelines to Clinical Management of Infectious Disease Preparing Committee, Urinary tract infection/male genital infection working group, et al. JAID/JSC guidelines for clinical management of infectious disease 2015—urinary tract infection/male genital infection. *J Infect Chemother* 2017; **23**: 733–51.
- Wagenlehner FM, Gasink LB, McGovern PC, et al; CERTAIN-1 Study Team. Cefepime-taniborbactam in complicated urinary tract infection. *N Engl J Med* 2024; **390**: 611–22.
- Portsmouth S, van Veenhuyzen D, Echols R, et al. Cefiderocol versus imipenem–cilastatin for the treatment of complicated urinary tract infections caused by Gram-negative uropathogens: a phase 2, randomised, double-blind, non-inferiority trial. *Lancet Infect Dis* 2018; **18**: 1319–28.
- Sims M, Mariyanovski V, McLeroth P, et al. Prospective, randomized, double-blind, phase 2 dose-ranging study comparing efficacy and safety of imipenem/cilastatin plus relebactam with imipenem/cilastatin alone in patients with complicated urinary tract infections. *J Antimicrob Chemother* 2017; **72**: 2616–26.

- 17 Wagenlehner FM, Sobel JD, Newell P, et al. Ceftazidime-avibactam versus doripenem for the treatment of complicated urinary tract infections, including acute pyelonephritis: RECAPTURE: a phase 3 randomized trial program. *Clin Infect Dis* 2016; **63**: 754–62.
- 18 Herrero FS. Ceftazidime–avibactam. *Rev Esp Quimioter* 2022; **25**: 40–42.
- 19 Ezure Y, Rico V, Paterson DL, et al. Efficacy and safety of carbapenems vs new antibiotics for treatment of adult patients with complicated urinary tract infections: a systematic review and meta-analysis. *Open Forum Infect Dis* 2020; **9**: ofaa480.
- 20 US Food and Drug Administration. Highlights of prescribing information, cefepime injection, for intravenous use. 2025. https://www.accessdata.fda.gov/drugsatfda_docs/label/2025/050817s012lbl.pdf (accessed Nov 11, 2025).
- 21 Fresenius-Kabi. AZTREONAM package insert. 2022. <https://products.fresenius-kabi.us/product-217.html> (accessed Sept 29, 2025).
- 22 Mallalieu NL, Winter E, Fettner S, et al. Safety and pharmacokinetic characterization of nacubactam, a novel β -lactamase inhibitor, alone and in combination with meropenem, in healthy volunteers. *Antimicrob Agents Chemother* 2020; **64**: e02229–19.
- 23 Qiu H, Wu T, Diao L, et al. Pharmacokinetics, safety, and tolerability of nacubactam (OP0595) after intravenous infusion: a randomized, double-blind, placebo-controlled phase I clinical trial in healthy Chinese male subjects. *Clin Exp Med* 2025; **26**: 96.
- 24 Pais GM, Change J, Barreto EF, et al. Clinical pharmacokinetics and pharmacodynamics of cefepime. *Clin Pharmacokinet* 2022; **61**: 929–53.
- 25 Nakashima M, Uematsu T, Kanamaru M, et al. Clinical phase I study of cefepime (BMJ-28142). *Chemotherapy* 1991; **39**: 104–16 (in Japanese).
- 26 Meyers BR, Wilkinson P, Mendelson MH, Bournazos C, Tejero C, Hirschman SZ. Pharmacokinetics of aztreonam in healthy elderly and young adult volunteers. *J Clin Pharmacol* 1993; **33**: 470–74.
- 27 Nakashima M, Uematsu T, Takiguchi Y, Maeda Y. Pharmacokinetics and safety of azthreonam in healthy Japanese volunteers. *Rinsho Yakuri* 1985; **16**: 409–16.